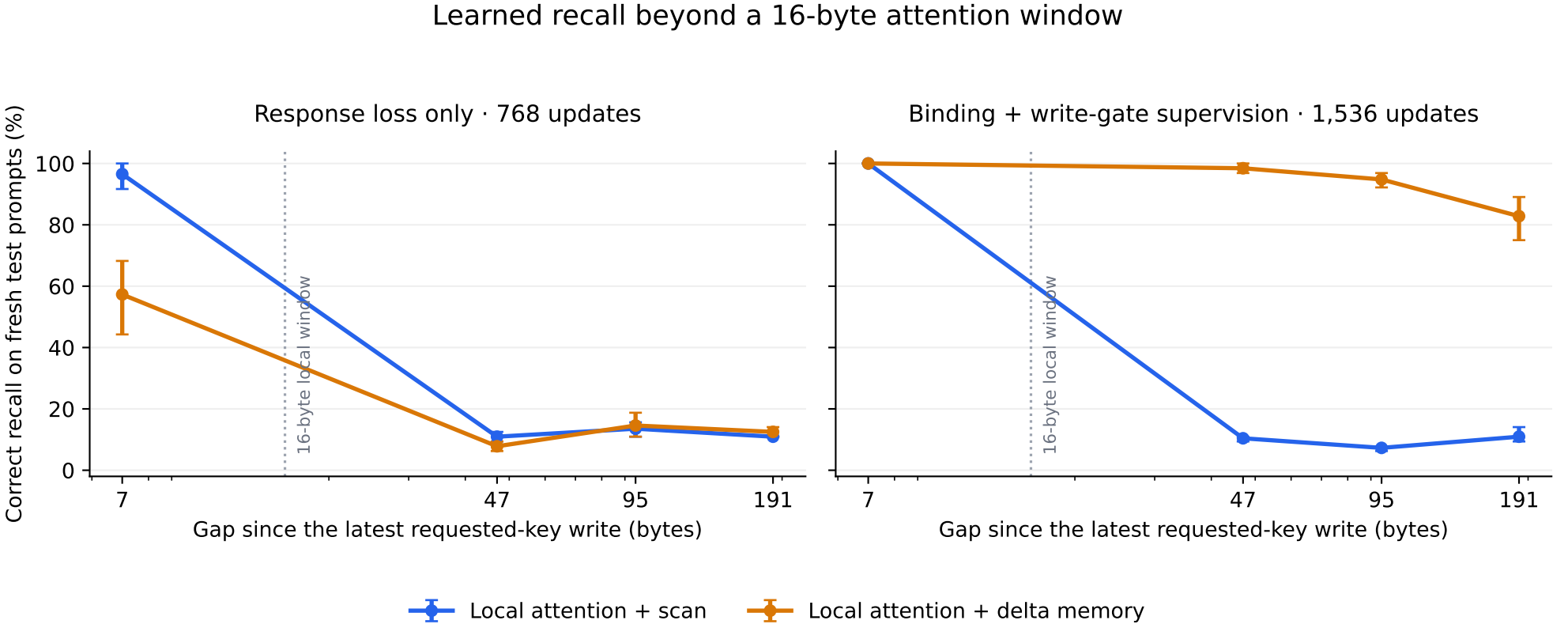
Learned recall beyond a 16-byte attention window
Response loss only · 768 updates
Binding + write-gate supervision · 1,536 updates
Correct recall on fresh test prompts (%)
100
80
60
40
20
0
16-byte local window
7
47
95
191
Gap since the latest requested-key write (bytes)
Local attention + scan
Local attention + delta memory
Means and seed ranges across three cold starts. Same grammar, randomized bindings; farthest test gap exceeds the 71-byte training maximum.